

4.1 Probability Distributions

1. Doctor's office collected the results of 350 patients who answered the following question: "Rate your pain from 1 to 10, where 1-almost no pain, 10-extreme pain". These were the results:

Rating, x	Frequency
1	22
2	27
3	35
4	36
5	40
6	56
7	70
8	31
9	13
10	20

2. Construct a probability distribution:

number of cars, x	households	$P(x)$
0	100	
1	325	
2	450	
3	94	
4	31	

Find the probability of randomly selecting a household that has at least two cars.
Find the mean (expected value).

3. Local charity is selling \$5 raffle tickets as part of Christmas fund-raising program. 1,200 tickets are sold. The four prizes are \$1,000, \$800, \$500, and \$350. You buy one ticket. What is $E(x)$, expected value of your gain/loss?

4. 1,500 students were asked: "How many parking tickets did you receive within the last 5 years?"

number of parking tickets x	$P(x)$	
0	0.150	
1	0.280	
2	0.267	
3	0.167	
4	0.100	
5	0.036	

4.2 Binomial Distributions

1. Decide whether the experiment is a binomial experiment. If it is, specify the values of n , p , q , and x . If it is not, explain why.

- a) A bag contains 4 white marbles, 7 red marbles, and 5 black marbles. You randomly select 3 marbles from the bag, without replacement. The random variable represents the number of black marbles.
- b) A certain medical procedure has a 85% chance of success on patients. The surgery is performed on 8 patients. The random variable represents the number of successful surgeries.
- c) You take an exam that consists of 10 multiple-choice questions. Each question has 5 possible answers, only one of which is correct. You will randomly guess the answer to each question. The random variable represents the number of correct answers.

2. Refer to 1b). Create a binomial probability distribution. Then, find the probability of the surgery being successful on a) exactly 5 patients, b) at least 6 patients, c) at most 3 patients.

3. Refer to 1c).

- a) Create a binomial probability distribution.
 - b) To get a B, you need to answer at least 8 questions correctly. Find the probability that you will get a B.
 - c) Find the probability that you answer at most 6 questions correctly.
 - d) Find the probability that you answer between 3 and 6 questions correctly.
4. 6.5% of people in the U.S. have A- blood type. You randomly select 6 Americans and ask them if their blood type is A-.
- a) Find the probability that all 6 are type A-.
 - b) Find the probability that at most 4 of them are type A-.